Esmolol

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Continuing Education Activity

Esmolol (esmolol hydrochloride) is an intravenous cardioselective beta-1 adrenergic antagonist. Esmolol is used in various settings, including urgent care, perioperatively, and postoperatively. It is indicated in sinus tachycardia, where a rapid rate requires intervention secondary to other comorbidities. Esmolol is FDA-approved for tachycardia and hypertension induced by intubation. Off-labeled it is used for rate and rhythm control in aortic dissection, acute coronary syndrome, non-ST elevation myocardial infarction, hypertensive emergencies, thyrotoxicosis, refractory ventricular tachycardia, refractory to defibrillation ventricular fibrillation, and to decrease catecholamine response during electroconvulsive therapy. This activity will highlight the mechanism of action, adverse events, pharmacology, monitoring, and relevant interactions of esmolol, pertinent for interprofessional team members in treating patients with tachycardia and other related conditions where esmolol will provide a benefit.

Objectives:

- Identify the mechanism of action of esmolol.
- Summarize the indications for using esmolol.
- Review the potential adverse effects of esmolol.
- Explain the importance of improving care coordination among the interprofessional team to enhance the delivery of care for patients when using esmolol.

Access free multiple choice questions on this topic.

Indications

Esmolol (esmolol hydrochloride) is an intravenous cardioselective beta-1 adrenergic antagonist.[1] In the contemporary era, esmolol has the following indications:

- Esmolol is FDA-approved for short-term use in controlling supraventricular tachycardia, such as a rapid ventricular rate in patients with atrial fibrillation or atrial flutter. It is also a useful emergency drug for focal atrial tachycardia, especially in patients with active bronchospasm.[2]
- Esmolol has also been reported to be a safe and effective drug for perioperative control of blood pressures, owing to its short half-life.[3]
- It is also indicated in sinus tachycardia, where a rapid rate requires emergent intervention, especially in acute coronary syndrome.[4]
- Esmolol is also recommended for the treatment of tachyarrhythmia in critically ill patients and hypertension induced by intubation.[5]
- Off-labeled, it is used in aortic dissection, acute coronary syndrome, non-ST elevation myocardial infarction, hypertensive emergencies, thyrotoxicosis, refractory ventricular tachycardia, refractory to defibrillation ventricular fibrillation.[6]
- It is also used to decrease catecholamine response during electroconvulsive therapy.[7]

Mechanism of Action

Esmolol is a short-acting, cardio-selective beta-blocker, a class II antiarrhythmic agent that is a competitive antagonist of the beta-1-adrenergic receptors primarily in the myocytes.[8] By blocking the adrenergic activity of epinephrine and norepinephrine, it decreases inotropic contractility, heart rate, and conduction.[9] Esmolol increases atrioventricular refractory time, decreases oxygen demand of the myocardium, and decreases atrioventricular conduction. A minor beta-2-adrenergic blockade has been reported with high intravenous infusion doses.

Esmolol is rapidly absorbed, the onset of action is within 60 seconds, and it maintains a steady-state within 5 minutes of initiation of infusion.[10] If a loading dose is administered, a steady-state can be achieved by the 2-minute mark. The drug has a 9-minute half-life and rapid renal clearance. Fast metabolism translates into the rapid decrease of pharmacological effect (within 10 to 30 minutes) once the infusion is discontinued, making esmolol safer when titrated appropriately.[11] Hydrolysis of the ester linkage metabolizes Esmolol by the esterases in the cytosol of erythrocytes. The metabolite has a half-life of 3.7 hours and is excreted via urine. Of note, esmolol does not have any membrane-stabilizing activity or alpha-adrenergic blockade.

Administration

Esmolol is administered intravenously, preferably through central venous access; however, it can also be infused peripherally. FDA approval was based on adult subjects; pediatric dosages have been experimental and are not approved by the FDA.

Perioperative and Postoperative Tachycardia and Hypertension [12]

Two Methods

- A bolus of 1000 mcg/kg over 30 seconds, followed by 150 mcg/kg per minute infusion, with a max dose of 300 mcg/kg per minute.
- A bolus of 500 mcg/kg over 1 minute, followed by 50 mcg/kg per minute infusion for 4 minutes. If the desired effect is not reached, it may increase in 50 mcg/kg per minute increments until the max dose of 300 mcg/kg per minute.

Supraventricular Tachycardia and Sinus Tachycardia [13]

- Bolus loading dose of 500 mcg/kg over 1 minute, followed by 50 mcg/kg/min infusion for 4 minutes. If the desired effect is not reached, it may increase in 50 mcg/kg per minute increments until the max dose of 200 mcg/kg per minute.
- For rapid efficacy, follow the initial loading dose and infusion of 50 mcg/kg per minute by a second 500 mcg/kg bolus over 1 minute and increase drip to 100 mcg/kg per minute for 4 minutes.

Off-label Use

Hypertensive Emergency [14] [15]

- Loading dose of 500 to 1000 mcg/kg over 1 minute (can be repeated once), followed by infusion of 50 mcg/kg per minute until the max dose of 200 mcg/kg per minute is achieved.

Aortic Dissection [16]

- Loading dose of 500 mcg/kg over 2 to 5 minutes, followed by infusion of 10 to 20 mcg/kg per minute.

Acute Coronary Syndrome [17]

- Loading dose of 500 mcg/kg over 1 minute, followed by 50 mcg/kg per minute infusion titrated every 5 to 15 minutes until the max dose of 300 mcg/kg per minute.

Electroconvulsive Therapy [18]

- Loading dose of 1000 mcg/kg over 1 minute before administration of anesthesia.

Thyrotoxicosis

- 50 to 100 mcg/kg per minute

Intubation Cholinergic Response

- Loading dose of 1000 to 2000 mcg/kg administered 1.5 to 3 min before endotracheal intubation.

Ventricular Tachycardia/Ventricular Fibrillation

- Loading dose of 500 mcg/kg bolus, followed by 50 mcg/kg per minute

Esmolol should be used short-term, with a maintenance dose infusion not exceeding 48 hours. Most patients who are on esmolol infusions will be transitioned to other long-term agents. Once the effects of esmolol are no longer needed, it should not be discontinued suddenly.[4] New medication should be initiated, followed by titrated esmolol down by 50% and reassessing hemodynamic stability. If the patient is stable, continue to slowly titrate the esmolol down while titrating the new agent up until the desired goal heart rate and/or blood pressure. Peripheral extravasations can cause thrombophlebitis. If the extravasation occurs, the infusion should be stopped, the line should be gently aspirated, and the limb should be elevated.

Specific Population

Pregnant Women: Esmolol is a category C drug in pregnancy. During trials in third-trimester pregnancy and labor, Esmolol has been shown to cause fetal bradycardia, which persisted after the drug infusion was discontinued.[19]

Breastfeeding Women: There is no conclusive data regarding milk excretion; however, it should not be used in breastfeeding mothers due to the serious potential for adverse effects.[20]

Renal or Hepatic Impairment: Esmolol does not require dosage adjustment for renal or hepatic impairment.[21][22]

Adverse Effects

- The most common adverse reaction with esmolol is a hemodynamic compromise, which happens to over 10% of all patients.[23]
- The risk of asymptomatic (25%) and symptomatic (12%) hypotension occurs at all doses, but the risk is dose-dependent.[24] Hypotension in Esmolol patients often appears once doses reach 150 mcg/kg per minute. In clinical trials, Esmolol-induced hypotension was usually corrected by titrating down or discontinuing the infusion.
- Patients also experienced dizziness (3%), peripheral ischemia (1%), and infusion site reaction (8%), such as blistering/necrosis/thrombophlebitis.[13]
- Rare side effects (less than 1% of patients without comorbidities) included bradycardia, decompensated heart failure, cardiac arrest, and heart block.[25]
- Esmolol did not show any significant laboratory abnormalities, and patients tolerated the drug well while being infused for up to 24 hours. However, some cases of hyperkalemia with esmolol use have been reported. Therefore, electrolyte levels should be monitored for hyperkalemia in patients who have renal disease.[26]
- Hypoglycemic-induced tachycardia is often not present with the use of Esmolol, like other beta-blockers. Patients with diabetes who are on Esmolol should be closely monitored. Esmolol has also been reported to exacerbate coronary vasospasms, such as Prinzmetal’s Angina.[27]
- In a setting of a patient with pheochromocytoma, esmolol should be given with an alpha-blocker not to have beta-blockage without opposed alpha. Patients with a history of hyperthyroidism should be closely monitored after discontinuation of esmolol may exacerbate hyperthyroidism. Esmolol may also aggravate arterial insufficiency in patients with a history of significant peripheral vascular disease.[28]

Contraindications

Esmolol is contraindicated in patients with sinus bradycardia, sick sinus syndrome, atrioventricular heart block, heart failure, cardiogenic shock, pulmonary hypertension, and a history of hypersensitivity reactions to esmolol. In addition, Esmolol and calcium channel blockers should not be given together, as this may exacerbate hypotension and bradycardia.

Patients with first-degree heart block and nodal dysfunctions are at an increased risk of progressive heart block, bradycardia, and AV dissociation. Patients with preexisting heart failure are at greater risk of decompensated heart failure and cardiogenic shock.[29]

Patients with airway disease, such as asthma and chronic obstructive pulmonary disease, should be cautious when using any beta-blocker, such as Esmolol.[30] Even though the effects of beta-2 is minimal, some risk may exist.

Drug-Drug Interactions

Numerous drug-drug interactions exist with esmolol, most significant being:[31][32][31]

- Digoxin: 10% to 20% increase of digoxin levels, with may aggravate slow AV conduction and heart rate
- Anticholinesterases: Prolonged neuromuscular junction blockage and recovery time
- Calcium channel blockers: Decreases myocardial contractility, may trigger a cardiac arrest
- Vasoconstrictor and inotropic agents: Esmolol should not be used to control heart rate elevation in patients who receive peripheral vascular constrictive agents such as epinephrine, norepinephrine, and dopamine.

Monitoring

- Esmolol is an ultra-fast-acting medication, which can alter heart rate and pressure very quickly. While the patient is on esmolol, it is recommended that they have continuous blood pressure monitoring, heart rate, mean arterial pressure, and, if possible, ECG.[33] An arterial line will provide the best method of continuous blood pressure monitoring, which is beneficial in use with esmolol.
- The intravenous (IV) site should be checked regularly when given peripherally to check any signs or symptoms of infiltration and extravasation.
- As mentioned above, in patients with comorbidities, appropriate blood levels should be monitored, such as glucose (diabetes), thyroid levels (hyperthyroid), potassium (renal insufficiency), and peripheral pulses (peripheral vascular disease).[34]

Toxicity

An overdose of esmolol can result in a myriad of symptoms and effects. Cardiac signs of toxicity include but are not limited to bradycardia, AV block of any degree, complete AV dissociation, decrease contractility, cardiogenic shock, asystole, and pulseless electrical activity.[35] Neurologic signs of toxicity include but are not limited to respiratory irregularities, seizures, coma, and psychiatric symptomatology.[36] Other symptoms of toxicity may include bronchospasms, gastrointestinal mesenteric ischemia, and peripheral cyanosis.

Since Esmolol has a very short half-life (9 minutes), toxicity should be treated by discontinuing the esmolol infusion. Acute toxicity is often self-limited and is treated supportively. Toxicity that results in bradycardia should be treated by atropine, pacing, and other anticholinergic agents. Cardiogenic shock can be treated with inotropic agents like dobutamine, dopamine, and isoproterenol. Bronchospasms, even though rare, should be treated with a beta-2 agonist, such as albuterol.[37] The Advanced Cardiac Life Support protocols should treat pulseless electrical activity and cardiac arrest. As with other beta-blocker overdoses, intravenous glucagon and calcium chloride can be given to mitigate the effects of esmolol while it is being metabolized.

Enhancing Healthcare Team Outcomes

Healthcare workers, including those in the ICU who administer esmolol, should be familiar with its adverse effects. Esmolol is an ultra-fast-acting medication, which can alter heart rate and pressure very quickly. Therefore, while the patient is on esmolol, it is recommended that nursing staff continuously monitor patients for blood pressure, heart rate, mean arterial pressure, and, if possible, ECG. An arterial line will provide the best method of continuous blood pressure monitoring, which is beneficial in use with esmolol. When given peripherally, the intravenous (IV) site should be checked regularly to ensure that it did not infiltrate and extravasate. As mentioned above, in patients with comorbidities, appropriate blood levels should be monitored, such as glucose (diabetes), thyroid levels (hyperthyroid), potassium (renal insufficiency), and peripheral pulses (peripheral vascular disease). The pharmacist should perform medication reconciliation and inform the prescriber of any concerns.

Given the pharmacodynamics of esmolol, it is advisable to use an interprofessional team approach to using the drug. This interprofessional team will include clinicians, specialists, mid-level practitioners, nurses, and pharmacists, all coordinating their activities and communicating openly so all team members are operating from the same information and can intervene or report the need for intervention and achieve optimal patient outcomes. [Level 5]